

PAPER I

1. $y = 2x - 5$

2. $(x-1)(x+2)(x-5)$

3. (a) $\begin{pmatrix} 8 \\ -5 \\ -5 \end{pmatrix}$ (b) 1 (c) 5 units

4. $3x + 4y = 26$

5. (a) $\vec{AB} = \begin{pmatrix} 6 \\ 3 \\ -6 \end{pmatrix}$ $\vec{AC} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ (b) $e(5, 0, 0)$ $d(7, 1, -2)$

6. (a) $x^4 + 4x^2 + 3$ (b) $(x^2+1)(x^2+3)$ 7. (a) $\frac{24}{25}$ (b) $\frac{7}{25}$ (c) $\frac{323}{325}$

8. (a) $u_n \rightarrow \text{LIMIT AS } n \rightarrow \infty$ since 0.3 lies between -1 and 1 (b) $5\frac{5}{7}$

(c) (i) $[u_0 = 1, u_1 = 2.6, u_2 = 7.4, u_3 = 21.8 \dots \text{(CONTINUE)} \dots u_7 = 1749.8]$

$\therefore$ SMALLEST VALUE OF n IS 7.

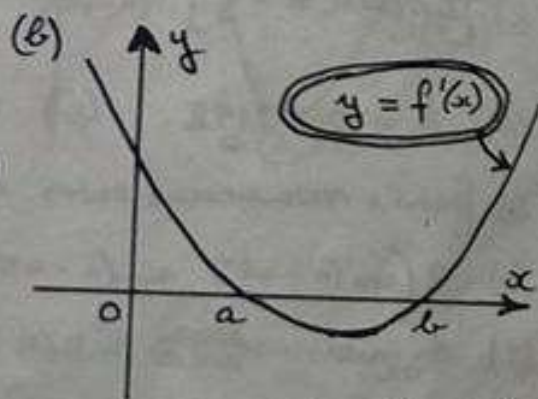
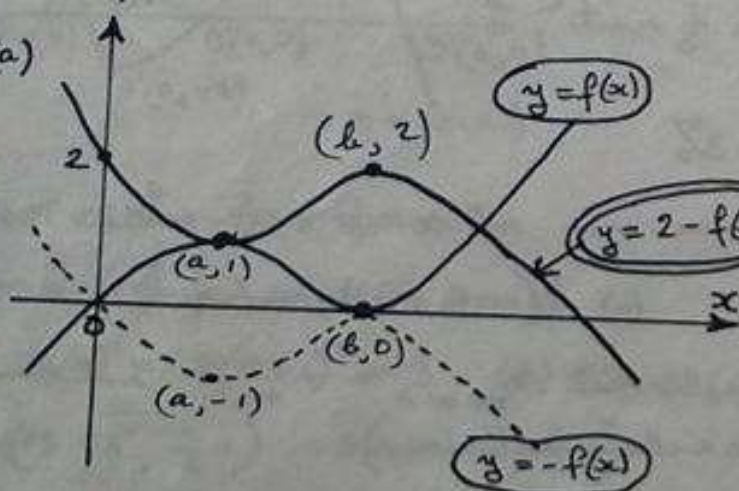
(ii) VALUE OF TERM = 1749.8.

9. $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}$

10. $y = 2x^3 - x^2 + 5$ 11. $k = 1$

12. 9.7

13. (a)



(c) $(0, \frac{1}{2})$

14. $3x^{\frac{1}{2}} + \frac{2}{x^{\frac{1}{2}}}$

15. (a) $\angle = 90^\circ$ $\angle = 135^\circ$ (b) $\frac{1}{2} \text{ unit}^2$

16. $\frac{3\sqrt{3}}{2}$

17. (a) 9.8 ms^{-1} (b) 2 seconds

18. (a) $\cos^2 \theta + 4 \cos \theta + 4 = 0$; Then prove that $\Delta = 0$

Since $\Delta = 0$, there are equal roots

(b) Prove that $\cos \theta = -2$ $\therefore$ NO SOLUTION.

19. $x = 1.544$

PAPER II

1. (a) $\vec{AB} = \begin{pmatrix} 1 \\ 7 \\ 2 \end{pmatrix}$ $\vec{AC} = \begin{pmatrix} 4 \\ 7 \\ -5 \end{pmatrix}$ (b) 51.9° (c) 27.43 units^2

2. (a) AND (b) $(0, -2)$ is a point of inflection
 $(3, 25)$ is a maximum S.P

(a) $2x + y = -1$ (b) 26.6°

4. (a) $a = 1, b = 6$ (b) 36 units^2 (c) (i) $P(5, 5)$ (ii) $20\frac{5}{6} \text{ units}^2$

5. (a) (i) $A(-2, 10)$ (ii) $y = 3x + 16$ (b) $B(4, 28)$

6. (a) $(x - \frac{1}{2})^2 + (y - 3)^2 = \frac{13}{4}$ (b) (i) $B(8, 8)$ (ii) $F(14, 12)$ $C(6\frac{1}{2}, 7)$

(c) WORKING OUT TO LEAD TO:- $A = \pi \frac{\sqrt{13}}{2} + \pi(2\sqrt{13}) + \pi(\frac{5\sqrt{13}}{2})$
etc

7. (a) $f(x) = \sqrt{13} \cos(x + 56.3^\circ)$ $h = \sqrt{13}$ $\alpha = 56.3^\circ$

(b) $\text{MAX} = \sqrt{13}$ occurs when $x = 303.7^\circ$

$\text{MIN} = -\sqrt{13}$ occurs when $x = 123.7^\circ$

(c) 0

8. (a) $1^{\text{st}} \text{ feed} = 1\text{g}$ $2^{\text{nd}} \text{ feed} = 1.75\text{g}$ $3^{\text{rd}} \text{ feed} = 2.3125\text{g}$ $4^{\text{th}} \text{ feed} = 2.7344\text{g}$
But $\frac{3}{4}$ of $2.3125 = 1.7344$ $\therefore$ 4 feeds are required

(b) (i) $u_{n+1} = 0.75u_n + 1$ ($u_0 = 1$) where u_n is the amount of plant food after n feeds.

(ii) Prove that the convergent value is 4

Since convergent value (4) $< 5\text{g}$, it can be fed indefinitely.

9. (a) $\frac{31\pi}{5} \text{ units}^3$ (b) (i) $\frac{16\pi}{3} \text{ units}^3$ (ii) $\frac{32\pi}{3} \text{ units}^3$

10. (a) [PROOF]

(b) S.A. is minimised when $r = \sqrt[3]{\frac{400}{3\pi}}$

TABLE OF VALUES
REQUIRED

11. (a) PROVE that $\log_e y = bx + \log_e a$ [cf. $y = mx + c$]
 $\therefore$ There is a linear relationship between $\log_e y$ and x .

(b) FIRSTLY :-

x	3.1	3.5	4.1	5.2
$\log_e y$	9.99	11.19	12.99	16.29

Use $1^{\text{st}}/3^{\text{rd}}$ points to obtain

$a = 1.99372$

$b = 3$

[using gradient and
 $\log_e y = 3x + \log_e a$]

$y = 1.99372e^{3x}$